

Chapter 7

TAXONOMY OF ORGANISMS

Brainstorming Questions:

- ✓ What is Taxonomy?
- ✓ How diversified organisms the world are managed to be studied?
- ✓ Explain the differences between classification, nomenclature and identification

Etymology of Taxonomy

Def. Taxonomy is the theory and practice of classification of organisms.

✓ The term taxonomy was coined by a Swiss naturalist, De Candolle, in the French form as “taxonomie” in 1813.

➤ Taxonomy is derived from two Greek words:

Taxis = to classify, to arrange;

nomos = law, science

❖ Thus, taxonomy means the law governing the classification or arrangement of organisms into taxa.

❖ some naturalists use the term “Taxonomy” as synonymous with “systematics”, but many naturalists consider taxonomy as one part of systematics, and it is mainly concerned with: Classification, Nomenclature and Identification

Thus, **systematics** is defined as

- the study and description of the diversities of organisms,
- the investigation of the causes and consequences of the variation, and
- the manipulation of the data obtained to produce a **system of classification, or**
- the scientific study of the kinds and diversity of organisms and of any and all relationships between them.

- i. Classification:** Is ordering/assignment of organisms into hierarchy of ranks or categories distinguished by certain characters; or based on similarities and/or differences.
- ii. Nomenclature:** Is the naming of groups of organisms and the rules governing the application of the names.
- iii. Identification:** Is the naming of an organism or a specimen by reference to an already existent classification. It involves referring an individual specimen to a previously classified and named system.

The need for classification is:

- to organize species into groups and to gather information about organisms in the world wide,
 - to identify new organisms and
 - to show relationships between organisms.
- ❖ Thus, **identification** follows **classification** and **nomenclature**, which means that classification and Nomenclature precede identification.

7.1. Early Attempts to Classify Organisms

Brainstorming questions:

1. How taxonomy is growing as a science?
2. What are the specific contributions of early naturalists for the development of modern taxonomy?

- Taxonomy was started as old as man himself.
- Among several early Greek and Roman philosophers who enumerated organisms, **Aristotle (384–322 BC)**, **Theophrastus (370-285 BC)**, Carolus Linnaeus (1707-1778), **M. Adanson (1727-1806)** and **Lamarck** to mention few.

- **Aristotle:** was a Greek philosopher who contributed most extensively to the development of the modern biology.
- ✓ He viewed life was hierarchical and he classified organisms, mainly history of Animals.
- **Theophrastus:** was a keen observer and was the first taxonomist to write down a classification in a permanent and logical form. For this reason, he was called the father of Botany by Linnaeus.
- ✓ As to him, any given organisms is either A or not A.
- ✓ According to his classification system, there are no any intermediate organisms.
- **Pliny the Elder (23-79 A.D.):** mentioned and categorized plants in terms horticultural practices, medicinal uses, and timber use practices.
- ✓ In his classification, he used a similar classification as his predecessors' errors.

- **Dioscorides (1st Century AD):** He was a Greek physician who studied and described about 600 medicinal plants in his book “De Materia Medica” with their useful applications.
- **Caesalpinus (1519-1603):** He classified plants based on **definite morphological criteria** mainly on the basis of **habit, fruits, seeds, and ovaries and their position.**
- **Linnaeus and Post Linnaean Classifications** Linnaeus produced important books that he classified all known animals, plants and minerals.
- ✓ Linnaeus was the **father of modern taxonomy**, and his classification system was popular largely because its simplicity and was artificial.

- **Adanson** is most remembered for championing the idea that in classification one should use a great range of characters covering all aspects of the organisms.
- **Lamarck** is best known for his theory of evolution, Lamarckism, whereby characters acquired during life become inherited.
- The evolutionary theory of **Darwin** had little immediate impact on classifications of organisms.
- The previous classifications were based on morphological similarities or dissimilarities.
- However, the early classification systems were broad and vague as indicated by generic terms like animals, worms, sedges, grasses and trees, etcetera.

7.2. Modern Views of Classification (Schemes of Classification)

1. What are the modern views of classification?
2. How the modern views of classification are based up on?

- Unlike the early classifications the modern classification system of diversified organisms is depend up on the combined evidence of morphology, ultrastructure and molecular phylogeny.
- In classification, though it is organisms that are being classified, their characters /character states are used to classify or categorize organisms.
- The differences in approach to comparison of character states are lead to the development of **five different** classification systems.

- **7.2.1. Artificial classification system**

- ✓ Artificial system was the first to be used in the classification of organisms
- ✓ Theophrastus and Linnaeus are good example of artificial system of classification.
- ✓ only one or at most a few characters are selected for use in making comparison among organisms i.e. a few characters are involved.
- ✓ Artificial system is monothetic i.e. possession of unique set of characters/features and not difficult

- **7.2.2. Natural classification system**

- ✓ is based up on several to many characters
- ✓ is therefore polythetic i.e. it places together organisms that have the greatest number of shared features,
- ✓ Adanson and De Jussieu are good examples of natural classification system.

7.2.3. Phyletic or Evolutionary classification system

- It is a classification that aims at reconstructing a sequence starting with the most primitive character and ending with the most advanced or derived character.
- It ensures that each taxon recognized as a **monophyletic** i.e. has arisen by the diversification of a single ancestor.

7.2.4. Phenetics classification system

- This is a classification where similarity/resemblance is determined based on a set of phenotypic characteristics of organisms.



7.2.5. Phylogenetic or cladistics classification system

- It is the concepts and methods of determining similarity by constructing branching patterns of evolution.
- Once the cladogram or tree of life is generated, it is possible to construct classification based upon the cladogram.

7.3. Domains of Life and the Hierarchical System of Classification

1. What are the major domains of life?
2. How organisms are arranged in the hierarchical classification?
3. Discuss about the different kingdom concepts, and which organisms are categorized in each kingdom?

- Linnaeus developed a hierarchical system of classification.
- An arrangement of taxa into an ascending series of ever-increasing inclusiveness forms is known as a hierarchical system of classification.
- The three domains are identified and organized based on the difference between eukaryotes and prokaryotes (Domain Archae bacteria, Eubacteria and Eukarya).

Diagrammatical representation of mammal and insect and hierarchical classification

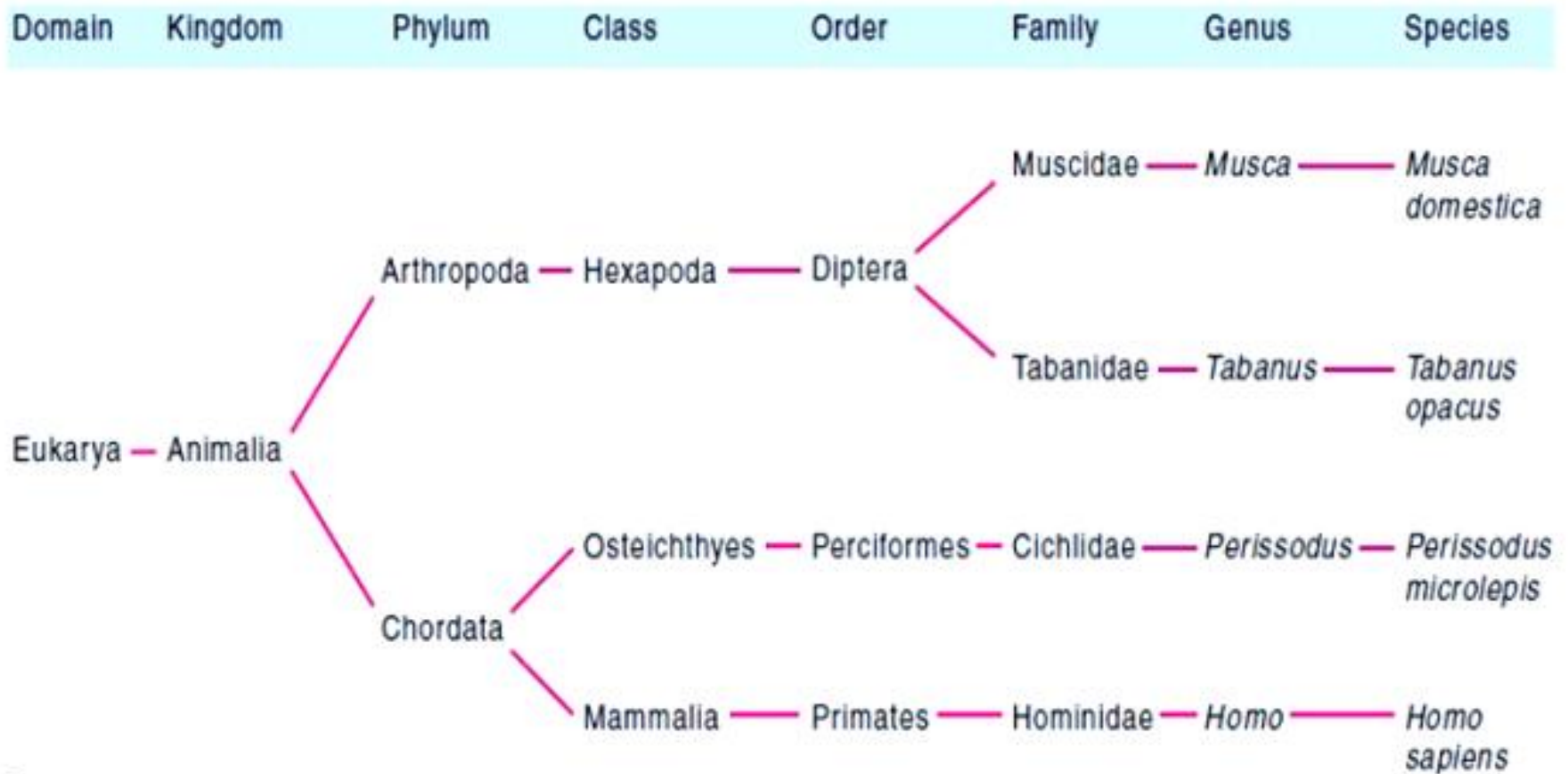


Fig: 7.2. Classification hierarchy of organism

In a hierarchical system, we start at the bottom with individuals (species) and end up at the top with one all-embracing taxon (Domain).

The major levels in the hierarchy are the following

Domain

Kingdom

Division/Phylum (pl. phyla)

Class

Order

Family

Genus (plural, genera)

Species

- Species are a series of recognizably similar individuals distinct from other species;
- they are the fundamental bases of the hierarchy of classification.
- Related species are grouped in to one genus.
- Related genera are grouped in to one family.....
- Related orders put in one class, classes in to division (if plant) and in to phylum (if animal) and in to kingdom and domain.

7.3.1. Concepts of the Kingdom

- In biology, **Kingdom** (Latin: **regnum**, pl. **regna**) is a taxonomic rank, which is the highest rank or in the more recent three-domain system, it is the second rank below the **Domain**.

i) The Two- Kingdom Concept/Approach

- ✓ The two kingdom concept existed before the 1950s.
- ✓ This concept was classifying organisms based up on the **cell wall present versus absent (Kingdom Plantae and Animalia)**

ii) Three-Kingdom Concept

- ✓ This kingdom concept/approach classifies all organisms in to three Kingdoms (kingdom Plantae, Animalia and Protista)
- ✓ (Kingdom Protista is for the intermediates, like some algae and protozoa, which behave like both plants and animals)

iii) The Four-Kingdom Concept

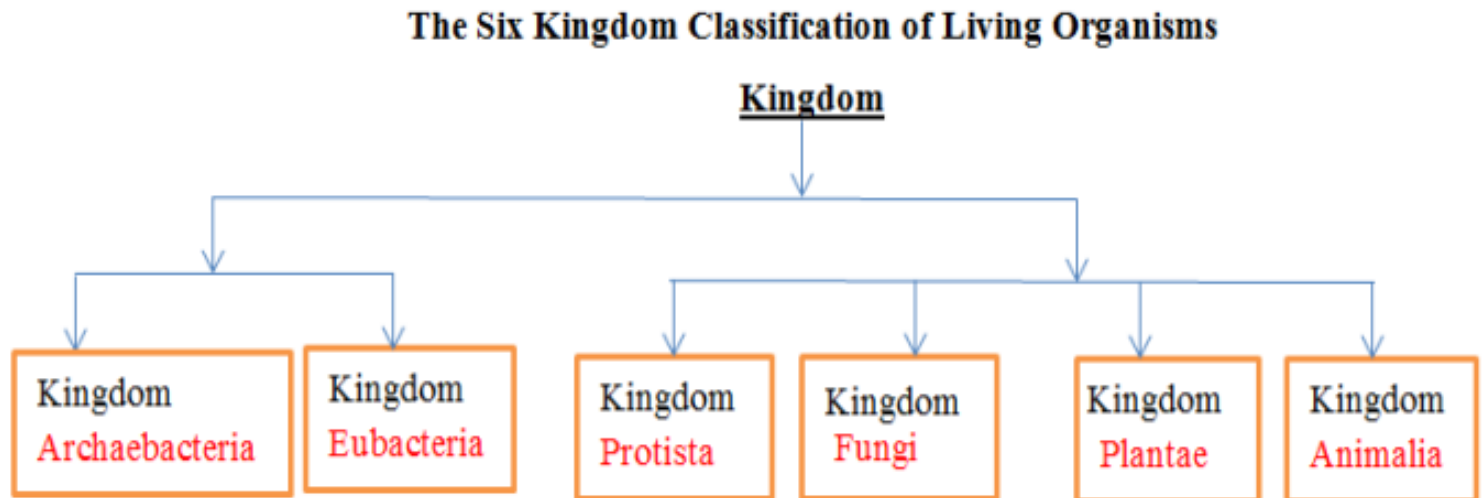
- ✓ Kingdom Plantae, Animal, Protista and Fungi. Fungi are organisms that do have both plant and animal characteristics.

iv) The Five-Kingdom Concept

- After the invention of the Transmission Electron Microscope (TEM), some biologists were able to investigate the composition of the cell.
- The five kingdoms are Kingdom Monera, kingdom Fungi, kingdom Protista, kingdom Plantae and kingdom Animalia

vi) The Six-Kingdom Concept

- This concept classifies first kingdom monera in to two groups based on the nucleus and other organelles have or lacking a membrane.



7.4. Binomial Nomenclature

1. What is the binomial nomenclature?
2. How the binomial nomenclature is applicable for organisms?
3. What are the preconditions for giving binomial nomenclature for organisms?
4. What is the purpose of giving scientific names for organisms?
5. What are characteristics scientific names?
6. What are operative principles of nomenclature?
7. How authorities of scientific names are cited?

Binomial nomenclature

- A binomial name is comprised of two parts, i.e. the generic name (genus name) and the specific name.
Synonyms: binominal nomenclature; binary nomenclature; two-term naming system.
- Nomenclature is the naming of organisms and the rules governing the application of these names.
- The rules of nomenclature begin from Linnaeus's of classification where he established the real beginning of a perfect nomenclature of organisms.

Rules to follow when using binomial nomenclatures are:

- ✓ The entire two-part name must be italicized or underlined if handwritten.
- ✓ The genus name is always written first.
- ✓ The first letter of the genus name is always capitalized.
- ✓ The species name is never capitalized

7.4.1.The Purpose of giving names to organisms

- Name is a conventional symbol or code that serves as a means of reference and avoids the need for continuous use of an inconvenient descriptive phrase
- It is used for act as vehicle of communication, to universalize the organism and to avoid any names.
- The Code of nomenclature tries to avoid the defects of vernacular names and for this reason sets of rules have been drawn up.
- Scientific names are written in Latin form as well as in Latin alphabet.

7.4.2. International Code nomenclature

- The code is a rulebook that contains principles, rules and recommendations which form the basis of the system of nomenclature.

7.4.3. International Organizations & Unions for the Stabilization of Changes

- ICBN: International Botanical Congress;
- ICZN: International Congress for Zoological Nomenclature.
- ICTF : The International Commission for the Taxonomy of Fungi

7.4.4. Names of taxa (the rank of the genus and above)

- The family names consist of one term only and called uninominal names. Family names are written with a capital initial letter. E.g. Asteraceae, Lamiaceae.
- Names of the genera are uninomial and singular nouns and written with capital initial letter. E.g. *Hagenia*, *Datura*
- Names of the species consist of two terms, and called binomial